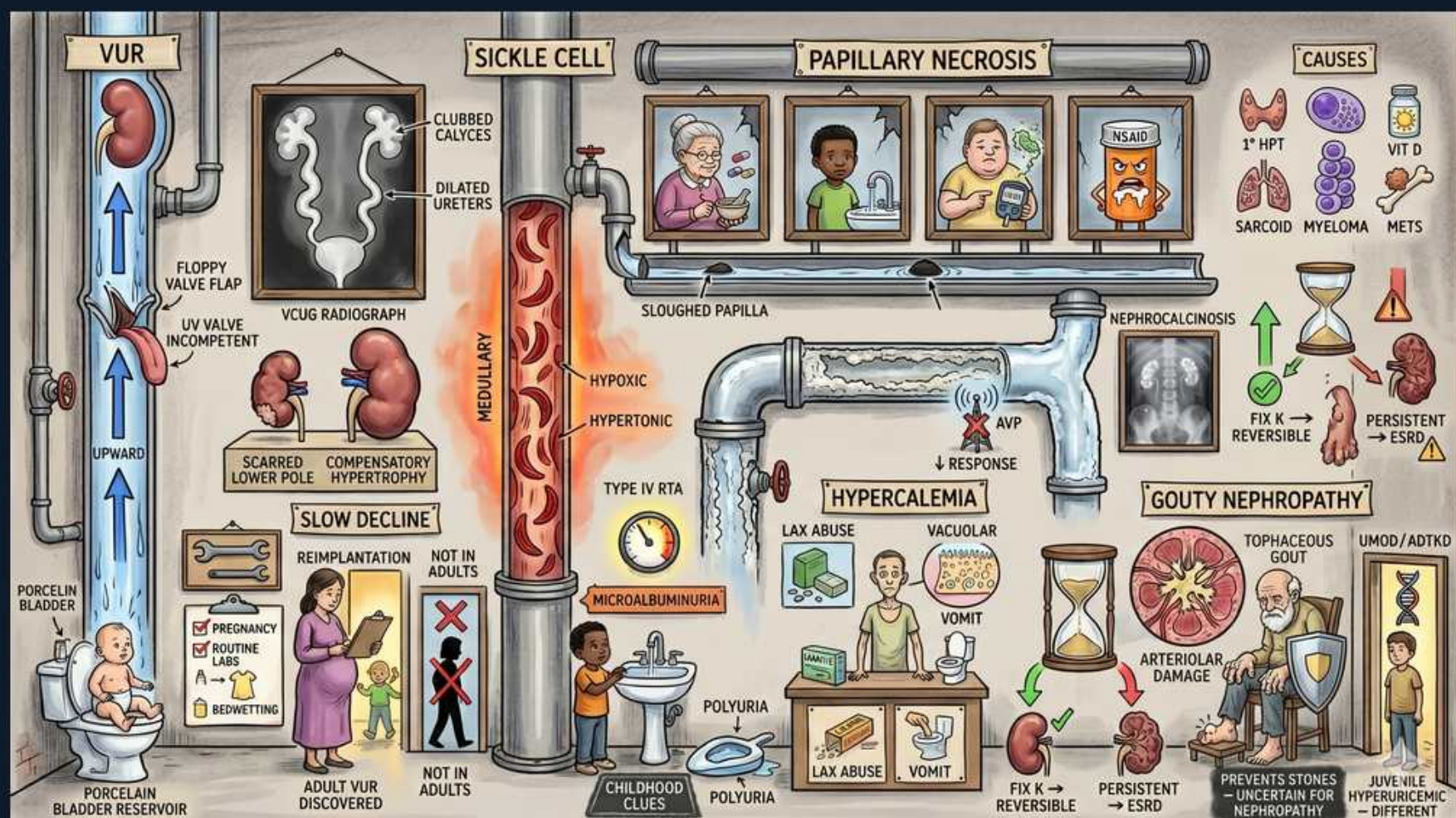


Tubulointerstitial — Structural/metabolic chronic CIN



1. VUR — vesicoureteral reflux

WHAT YOU SEE

A VUR banner with blue arrows flowing UPWARD past a floppy valve flap at the bladder — the urine running the wrong way up the ureter IS retrograde reflux through an incompetent ureterovesical valve.

WHAT IT ENCODES

Vesicoureteral reflux is retrograde flow of urine from bladder up the ureter due to an incompetent (short/abnormal) ureterovesical junction valve; in children it allows infected urine to reach the kidney, causing recurrent pyelonephritis and chronic scarring = reflux nephropathy, a leading cause of childhood CKD/hypertension. Graded I–V by severity. Many low-grade cases resolve with growth; high-grade or breakthrough infections may need prophylactic antibiotics or surgical reimplantation.

BOARD TRAP

Reflux nephropathy/CKD is presumed to start fresh in an adult who has 'just developed reflux.' Wrong — reflux nephropathy is largely a disease of EARLY CHILDHOOD; the scarring happens in young kidneys and presents later as adult hypertension/CKD/proteinuria. Onset in childhood (not new adult reflux) is the discriminator — adults present with the sequelae, not new scarring.

2. VCUG diagnosis

WHAT YOU SEE

A VCUG radiograph showing dilated ureters and blunted, clubbed calyces — the contrast tracking up dilated ureters with clubbed calyces IS the imaging proof of reflux and its cortical scars.

WHAT IT ENCODES

Voiding cystourethrogram (VCUG) is the diagnostic test for VUR: contrast is instilled in the bladder and fluoroscopy during voiding shows retrograde reflux up dilated ureters and grades it. Established reflux nephropathy shows cortical scars overlying CLUBBED (blunted/dilated) calyces, classically at the upper and lower poles, on imaging. DMSA renal scan detects cortical scars/differential function.

BOARD TRAP

Hydronephrosis with clubbed calyces is attributed to 'current obstruction, place a stent/nephrostomy.' Wrong — clubbed calyces with cortical scars and a history of childhood UTIs reflect old reflux nephropathy, not acute obstruction; a VCUG (not decompression) is the relevant test. Cortical scarring over clubbed calyces (chronic) vs acute hydronephrosis is the discriminator.

3. Scarred lower pole + compensatory hypertrophy

WHAT YOU SEE

Two kidneys, one with a SCARRED LOWER POLE and the other showing COMPENSATORY HYPERTROPHY — the shrunken scarred pole beside the enlarged partner IS focal reflux scarring driving hyperfiltration of the remaining nephrons.

WHAT IT ENCODES

Reflux nephropathy produces focal cortical scars (classically upper and lower poles where compound papillae permit intrarenal reflux), giving an irregular, asymmetric small kidney; unaffected nephrons and the contralateral kidney undergo compensatory hypertrophy and hyperfiltration. That hyperfiltration eventually causes secondary FSGS, proteinuria and progressive CKD even after infections stop. Hypertension is common.

BOARD TRAP

New proteinuria in a reflux-nephropathy patient is read as 'a separate primary glomerular disease.' Wrong — the proteinuria is usually secondary FSGS from hyperfiltration of remaining nephrons; ACEi/ARB to reduce intraglomerular pressure is the treatment, not immunosuppression. Secondary (hyperfiltration) vs primary FSGS is the discriminator and changes management.

4. Papillary necrosis

WHAT YOU SEE

A PAPILLARY NECROSIS banner over framed patient faces (elderly, sickle-cell, diabetic) and an angry NSAID bottle — the gallery of risk-factor faces plus NSAID IS the high-yield causes of papillae infarcting.

WHAT IT ENCODES

Renal papillary necrosis is ischemic/toxic death of the renal papillae (the medulla's blood supply is marginal). Mnemonic 'POSTCARDS' or the high-yield five: Pyelonephritis (severe/obstructive), Obstruction, Sickle cell disease/trait, Tuberculosis, Analgesics (phenacetin/NSAIDs), plus Diabetes mellitus and Cirrhosis. Presents with gross hematuria, flank/colicky pain (sloughed papilla acts like a stone), and possible sterile pyuria; CT urogram shows the classic 'ring shadow'/blunted papillae.

BOARD TRAP

Colicky flank pain with hematuria and a filling defect is automatically called a 'ureteral calculus.' Wrong — in a diabetic, sickle-cell, or chronic-analgesic patient the obstructing 'stone' may be a SLOUGHED NECROTIC PAPILLA; CT urogram showing papillary defects/ring shadow distinguishes it. The risk-factor background plus papillary imaging is the discriminator from a true calcium stone.

5. Sloughed papilla

WHAT YOU SEE

A sloughed papilla lodged inside a pipe drawn as HYPOXIC and HYPERTONIC medulla — the chunk of dead tissue stuck in the collecting pipe IS a necrotic papilla obstructing like a stone and causing hematuria.

WHAT IT ENCODES

The renal medulla operates at the edge of hypoxia (countercurrent exchange shunts oxygen away) and is hypertonic, so it is uniquely vulnerable; necrotic papillae detach and slough into the collecting system, where they cause hematuria and can obstruct the ureter. Vasoconstrictive insults (NSAIDs blocking medullary prostaglandins, sickling, dehydration) tip the medulla into infarction. The sloughed tissue may be passed in urine.

BOARD TRAP

Tissue passed in the urine with flank pain is dismissed as a 'blood clot.' Wrong — necrotic sloughed papilla can be passed and signals papillary necrosis (analgesic/sickle/diabetic medulla); recognizing it triggers a different workup than simple clot colic. Necrotic papillary tissue (vs clot) on inspection/pathology is the discriminator.

6. Analgesic/NSAID cause

WHAT YOU SEE

An NSAID bottle beside framed faces of chronic-pain patients — the pill bottle next to the long-suffering faces IS years of analgesic abuse driving medullary ischemia and papillary necrosis.

WHAT IT ENCODES

Chronic analgesic abuse (historically PHENACETIN-containing combination analgesics; today chronic NSAIDs, often with aspirin/caffeine) causes chronic interstitial nephritis and papillary necrosis. NSAIDs inhibit medullary vasodilatory prostaglandins, reducing medullary blood flow; phenacetin's metabolite concentrates in the medulla. Clues: middle-aged woman with chronic pain/headaches, analgesic dependence, sterile pyuria, and small bumpy kidneys; there is also an increased risk of urothelial carcinoma.

BOARD TRAP

Slow CKD with sterile pyuria in a chronic-pain patient is called 'idiopathic CKD' without a medication history. Wrong — a careful analgesic history (years of NSAID/combination analgesic use) reveals analgesic nephropathy, which is partly preventable by stopping the drugs; missing it forfeits intervention. The cumulative analgesic exposure history is the discriminator.

7. Type IV RTA / microalbuminuria

WHAT YOU SEE

A TYPE IV RTA dial and a MICROALBUMINURIA label on the sickle-cell medulla column — they ARE sickle nephropathy's hyperkalemic distal RTA and the early albuminuria from hyperfiltration.

WHAT IT ENCODES

Sickle cell disease/trait causes a medullary nephropathy: sickling in the hypoxic, hypertonic, acidic vasa recta → microinfarcts, impaired concentrating ability (hyposthenuria/nocturia), painless hematuria, papillary necrosis, a hyperkalemic type IV-like distal RTA, and progressive glomerular injury (hyperfiltration → FSGS → microalbuminuria/proteinuria, an early marker). ACEi/ARB are used for the albuminuria.

BOARD TRAP

Painless hematuria in a young Black patient with sickle TRAIT is assumed benign and not worked up, or proteinuria is ignored. Wrong — sickle nephropathy (even in trait) causes papillary necrosis, hematuria and progressive proteinuria/CKD that warrants ACEi/ARB and monitoring; also consider renal medullary carcinoma in sickle-trait patients with hematuria. Sickle status changes the differential — that is the discriminator.

8. Nephrocalcinosis

WHAT YOU SEE

A radiograph showing diffuse renal calcification (NEPHROCALCINOSIS) with a crossed-out AVP/response — the speckled calcified kidney IS calcium deposition that wrecks the concentrating (ADH) response.

WHAT IT ENCODES

Nephrocalcinosis = diffuse calcium deposition in the renal parenchyma (medullary pattern most common), driven by hypercalcemia/hypercalciuria, and causes chronic interstitial damage with an impaired AVP/concentrating response (acquired nephrogenic DI). Classic associations: distal (type I) RTA, primary hyperparathyroidism, sarcoidosis, medullary sponge kidney, and milk-alkali. It is visible on plain film/CT/ultrasound (medullary calcifications).

BOARD TRAP

Medullary nephrocalcinosis is paired with type II (proximal) RTA. Wrong — DISTAL (type I) RTA causes nephrocalcinosis and stones (alkaline urine, hypocitraturia, hypercalciuria), whereas proximal (type II) RTA does NOT cause nephrocalcinosis. The RTA type (I causes it, II does not) is the discriminator.

9. Hypercalcemia causes

WHAT YOU SEE

A CAUSES panel listing 1° HPT, SARCOID, MYELOMA, METS and VIT D around drawn glands/lungs/bone — these icons ARE the high-yield sources of the hypercalcemia that calcifies and scars the kidney.

WHAT IT ENCODES

Chronic hypercalcemia/hypercalciuria injures the kidney via vasoconstriction, nephrogenic DI (calcium downregulates aquaporins via the calcium-sensing receptor → polyuria), nephrocalcinosis, stones, and chronic interstitial nephritis. High-yield causes: primary hyperparathyroidism, sarcoidosis (excess 1,25-vitamin D), multiple myeloma, bone metastases/malignancy (PTHrP), and vitamin D toxicity. The earliest renal effect is often impaired concentrating ability with polyuria/polydipsia.

BOARD TRAP

Polyuria with hypercalcemia is labeled 'central diabetes insipidus.' Wrong — hypercalcemia causes NEPHROGENIC DI (kidney can't respond to ADH), so urine fails to concentrate after desmopressin; correcting the calcium reverses it. Nephrogenic (no DDAVP response) vs central (DDAVP responds) is the discriminator.

10. Hypokalemia — lax/vomit

WHAT YOU SEE

A wasting man vomiting beside LAX ABUSE and VACUOLAR labels — the laxatives, vomiting and vacuolated tubules ARE chronic potassium depletion causing reversible hypokalemic nephropathy.

WHAT IT ENCODES

Chronic hypokalemia (laxative abuse, recurrent vomiting, bulimia, diuretics, primary hyperaldosteronism) causes 'hypokalemic nephropathy': vacuolar degeneration of proximal tubular cells, impaired concentrating ability with nephrogenic DI (polyuria/polydipsia), increased ammoniogenesis, and chronic interstitial fibrosis with cyst formation if prolonged. Crucially it is REVERSIBLE if potassium is repleted before fibrosis sets in.

BOARD TRAP

Polyuria and CKD with low potassium are attributed to an irreversible structural disease and potassium is not corrected. Wrong — early hypokalemic nephropathy is reversible with potassium repletion and removing the cause (laxatives/vomiting); only chronic untreated injury becomes fixed fibrosis. Duration/reversibility (correct K early) is the discriminator and the actionable lever.

11. Polyuria from NDI

WHAT YOU SEE

A child standing at a sink labelled POLYURIA — the constant urine output IS the acquired nephrogenic DI that both low K⁺ and high Ca²⁺ cause by crippling the collecting duct's ADH response.

WHAT IT ENCODES

Both chronic hypercalcemia and chronic hypokalemia produce acquired nephrogenic diabetes insipidus by impairing the collecting duct's response to ADH (calcium acts on the CaSR; potassium depletion downregulates aquaporin-2), giving polyuria, nocturia and dilute urine. The defining feature is failure to concentrate the urine after desmopressin. Correcting the calcium or potassium early reverses the concentrating defect.

BOARD TRAP

Nephrogenic DI from electrolyte disturbance is treated with desmopressin alone. Wrong — the tubule cannot respond to ADH, so DDAVP fails; the fix is correcting the underlying calcium/potassium (and thiazide + low-salt/amiloride for persistent nephrogenic DI). Treating the metabolic cause, not giving more ADH, is the discriminator.

12. Gouty / urate nephropathy

WHAT YOU SEE

A GOUTY NEPHROPATHY panel with a tophaceous-gout foot and damaged arterioles — the chalky tophus plus arteriolar damage IS chronic urate deposition alongside the hypertension/vascular disease that really drives the CKD.

WHAT IT ENCODES

Chronic urate nephropathy: monosodium urate crystals deposit in the medullary interstitium causing chronic interstitial inflammation/fibrosis, often alongside tophaceous gout, hypertension and arteriolar/vascular disease. It is distinct from acute uric acid crystal nephropathy (TLS). Modern data suggest most CKD in gout patients is driven by comorbid hypertension, diabetes and vascular disease rather than urate deposition per se.

BOARD TRAP

Aggressive urate-lowering is assumed to clearly halt or reverse CKD progression in chronic gout. Wrong — trials have NOT consistently shown that lowering urate prevents renal progression in CKD; the renal decline is largely from coexisting hypertension/diabetes/vascular disease. Urate-lowering treats gout but is not a proven nephroprotective intervention — that nuance is the discriminator.

13. Reversible vs ESRD

WHAT YOU SEE

Twin panels: a green-check FIX K = REVERSIBLE versus PERSISTENT → ESRD with an hourglass — they ARE the therapeutic window: correct the electrolyte early and the kidney recovers, leave it and fibrosis becomes permanent.

WHAT IT ENCODES

Metabolic tubulointerstitial injuries (hypokalemic and hypercalcemic nephropathy) are REVERSIBLE if the electrolyte derangement is corrected early, before interstitial fibrosis and tubular atrophy become established; persistent uncorrected injury progresses to irreversible chronic interstitial fibrosis and ESRD. The therapeutic window is the message: identify and treat the metabolic driver promptly.

BOARD TRAP

A patient with hypokalemic/hypercalcemic nephropathy and a modestly elevated creatinine is told the kidney damage is permanent, so the electrolyte cause is not corrected. Wrong — early intervention can reverse the concentrating defect and halt progression; only long-standing fibrosis is fixed. Timing (early correctable vs late fibrotic) is the discriminator that determines reversibility.